4 **(a)** State the *First Law of Thermodynamics*.

.....

.....

..... [1]

(b) 0.050 moles of an ideal gas is contained within an uninsulated cylinder with a movable piston as shown in Fig. 4.1.

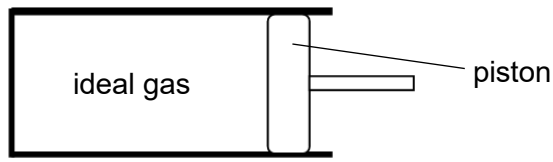

Fig. 4.1

The piston moves slowly outwards, resulting in the variation of pressure shown in Fig. 4.2

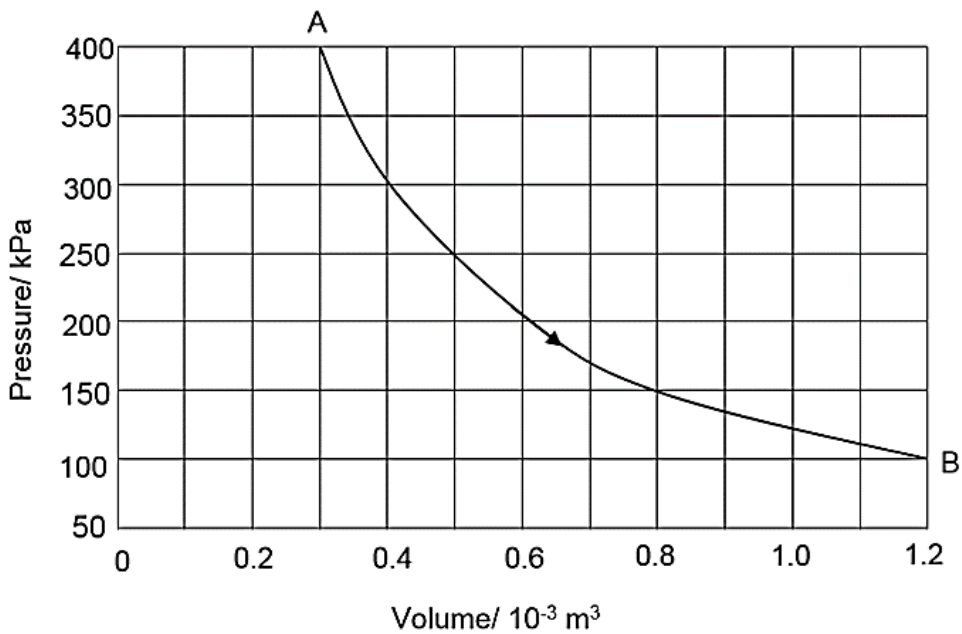

Fig. 4.2

(i) The process A to B occurs along an isotherm.

Determine the temperature along this isotherm.

temperature = ... K [2]

(ii) Calculate the total kinetic energy as a result of the random motion of the gas molecules in the cylinder.

total kinetic energy = ... J [1]

(iii) Estimate the amount of work done by gas as it expands from A to B

work done = ... J [2]

(iv) A student states that no heat flows into or out of the gas during the process A to B as the temperature of the gas did not change.

State and explain the validity of his statement.

.....

.....

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..... [3]

(c) In another set up with the same starting point as the one shown in Fig. 4.1, the piston is moved very quickly outwards expanding the gas to a volume of 1.2×10^{-3} m^3. As a result, the temperature of the gas decreases to 144 K.

On Fig. 4.2, sketch the variation with volume of the pressure of the gas. [2]